Huey's Home Lab Build Documentation

Cybersecurity Home Lab — Build Log & Troubleshooting Guide

1. Project Overview

This document details the complete build process of a cybersecurity home lab from scratch, including all errors encountered, troubleshooting steps taken, and final configurations. The lab consists of a Kali Linux attack machine and a Windows Server 2025 target machine connected through GNS3 for network simulation.

2. Host Machine Specifications

Component	Details
Processor	Intel Core i5-13500HX (13th Gen)
Cores / Logical Processors	14 Cores / 20 Logical Processors
RAM	16 GB
Storage	475 GB SSD (C: Drive)
Virtualization	Enabled in BIOS
OS	Windows 11

3. Software Installed

Software	Purpose	Status
VMware Workstation Pro 26H1	Desktop Hypervisor	Installed
Oracle VirtualBox	Desktop Hypervisor	Installed
GNS3 2.2.59	Network Simulator	Installed
GNS3 VM 2.2.59	GNS3 Backend Server	Installed in VMware
PuTTY 0.84	SSH/Telnet Client	Installed
WinSCP 6.5.6	File Transfer (SCP)	Installed
Wireshark	Packet Analyzer	Installed
Cisco Packet Tracer	Network Simulation	Installed

4. Virtual Machines

4.1 Huey's Enterprise (Kali Linux — Attack Machine)

Setting	Value
Original Name	Zee's Comp
Current Name	Huey's Enterprise
Hypervisor	Oracle VirtualBox
OS	Kali Linux 2024.3 (Debian 64-bit)
RAM	2048 MB
Static IP	192.168.99.20
Subnet Mask	255.255.255.0
Network	Inside (VMware VMnet — 192.168.99.0/24)
GNS3 Project	Huey's Starship
Default Login	kali / kali

4.2 Windows Server 2025 (Target Machine)

Setting	Value
Hypervisor	VMware Workstation Pro
OS	Windows Server 2025 Datacenter Evaluation
License	180-day Evaluation (No product key required)
RAM	4096 MB
Storage	60 GB (Single file)
CPU Cores	2
Static IP	192.168.99.10
Subnet Mask	255.255.255.0
Network Adapter	Inside (Host-only — VMnet 192.168.99.0/24)
Admin User	Tennis

5. Virtual Network Configuration (VMware)

Name	Type	Subnet	DHCP	Purpose
VMnet0	Bridged	Auto	No	External network access
VMnet1	Host-only	192.168.108.0	Yes	GNS3 VM network
Inside	Host-only	192.168.99.0	Yes	Lab internal network

DMZ	Host-only	192.168.247.0	Yes	DMZ network
VMnet8	NAT	192.168.214.0	Yes	Internet access for VMs

6. GNS3 Configuration

6.1 GNS3 VM Details

Setting	Value
GNS3 Version	2.2.59
VM Version	0.19.0
IP Address	192.168.108.128
Port	80
SSH Access	ssh gns3@192.168.108.128 (password: gns3)
Web UI	http://192.168.108.128 (admin/admin)
RAM Allocated	2048 MB
vCPUs	2
Project Name	Huey's Starship

7. Errors & Troubleshooting Log

Error 1: VMware Workstation Pro Download Issues (Broadcom Portal)

Problem: After creating a Broadcom account, the 'I agree to Terms and Conditions' checkbox was unclickable, and a 'Screening Required' message blocked the download.

Solution: Clicked the 'Free Software Downloads available HERE' link on the My Downloads page instead of downloading directly. This bypassed the screening requirement and provided access to the free VMware Workstation Pro download.

Error 2: GNS3 VM — Virtualized Intel VT-x/EPT Not Supported

Problem: When attempting to power on the GNS3 VM, VMware displayed: 'Virtualized Intel VT-x/EPT is not supported on this platform.'

Solution: Edited the GNS3 VM.vmx file directly using VS Code and made the following changes:

- Changed vhv.enable = "true" to vhv.enable = "false"
- Added hypervisor.cpuid.v0 = "FALSE" at the bottom of the file

This resolved the nested virtualization error and allowed the GNS3 VM to boot successfully.

Error 3: GNS3 VM — Failed to Allocate Main Memory

Problem: Error message: 'Could not create anonymous paging file for 4100 MB: The paging file is too small. Failed to allocate main memory.'

Solution: Reduced the GNS3 VM RAM from 4096 MB to 2048 MB in VMware settings. The host machine had insufficient free RAM due to Windows consuming approximately 10 GB of the 16 GB total.

Error 4: C: Drive Full (192 MB Free)

Problem: The C: drive had only 192 MB free out of 475 GB, causing VM disk errors and preventing lab operations.

Solution: Freed up approximately 145 GB by:

- Deleting duplicate Kali Linux VDI files (39 GB + 13.5 GB)
- Emptying the Recycle Bin
- Deleting temporary files (10.5 GB)
- Moving school and project folders to external drive

Result: C: drive went from 192 MB free to 145 GB free.

Error 5: VMware VIX API Not Found

Problem: GNS3 displayed: 'VMware vmrun tool could not be found. VMware or the VIX API is probably not installed.'

Solution: This was partially resolved by proceeding through the GNS3 setup wizard despite the warning. GNS3 was still able to detect and list the GNS3 VM after clicking Refresh in the VM name dropdown. The GNS3 VM must be started manually in VMware before opening GNS3 each session as a workaround.

Error 6: VM Renaming Issues (Zee's Comp to Huey's Enterprise)

Problem: Renaming the Kali Linux VM files outside of VirtualBox caused inaccessible VM errors, broken VDI references, and GNS3 showing the old name 'Zee's Comp' instead of 'Huey's Enterprise'.

Solution (multi-step):

- Removed the inaccessible VM from VirtualBox
- Renamed the folder from 'Zee_s Comp' to 'Huey's Enterprise' in File Explorer
- Renamed the VDI file to 'Huey's Enterprise.vdi'
- Edited the .vbox XML file in Notepad to remove the old 'Zee's Comp.vdi' HardDisk entry
- Re-added the VM to VirtualBox using Machine → Add and selecting the updated .vbox file
- Re-pointed the storage controller to the new Huey's Enterprise.vdi in VirtualBox Settings → Storage
- Re-added the VM template in GNS3 under VirtualBox VMs

Error 7: GNS3 Link Showing Red / NPF Error

Problem: After connecting Huey's Enterprise and Windows Server 2025 in GNS3, the link showed red with error: 'Could not find NPF id for interface VMnet0.'

Solution: Changed the Windows Server 2025 network adapter in VMware settings from VMnet0 (Bridged) to Inside (Host-only, 192.168.99.0/24). This matched the subnet being used for the lab's internal network. After restarting GNS3 and redrawing the link, both connections showed green.

Error 8: DHCP Overriding Static IP on Kali

Problem: Kali Linux kept receiving a DHCP-assigned IP (192.168.99.129/130) from VMware's Inside network DHCP server, which took priority over the manually set static IP of 192.168.99.20.

Solution:

- Removed the DHCP address: sudo ip addr del 192.168.99.129/24 dev eth0
- Configured /etc/network/interfaces with static IP settings
- Disabled NetworkManager management of eth0: sudo nmcli device set eth0 managed no
- Created permanent NetworkManager config: /etc/NetworkManager/conf.d/99-unmanaged-eth0.conf

Result: eth0 now permanently holds 192.168.99.20 without DHCP interference.

8. Final Network Configuration

Device	IP Address	Subnet	Network
Huey's Enterprise (Kali)	192.168.99.20	255.255.255.0	Inside
Windows Server 2025	192.168.99.10	255.255.255.0	Inside
GNS3 VM	192.168.108.128	255.255.255.0	VMnet1

9. Connectivity Verification

Ping test from Huey's Enterprise (Kali) to Windows Server 2025:

`Command: ping 192.168.99.10`

`Result: 9 packets transmitted, 9 received, 0% packet loss, avg time 1.055ms`

STATUS: SUCCESSFUL — Both machines communicating on the 192.168.99.0/24 network.

10. Session Startup Procedure

Follow this order every time you start a lab session:

1. Start GNS3 VM manually in VMware Workstation
2. Open GNS3 Windows application
3. Open the Huey's Starship project
4. Click the green Play button in GNS3
5. If Kali doesn't start automatically, power it on manually in VirtualBox
6. Verify eth0 on Kali shows 192.168.99.20 (run: ip addr show eth0)
7. Verify Windows Server shows 192.168.99.10 (run: ipconfig)

8. Run ping test to confirm connectivity

11. Next Steps

- Configure Active Directory on Windows Server 2025
- Set up DNS and DHCP roles on Windows Server
- Add more VMs to the GNS3 topology
- Practice offensive and defensive security scenarios between Kali and Windows Server
- Document attack and defense techniques
- Install VMware VIX API to enable full GNS3 control of VMware VMs
- Consider adding a router/switch between VMs in GNS3 for more realistic topology

Document prepared by Huey | Influential, LLC | Atlanta, GA